

Table 2. Molecular-model performance across the twelve Tox21 endpoints. Endpoint-specific AUPRC and macro-AUPRC are reported for the nuclear-receptor (NR) and stress-response (SR) groups. The best and second-best values in each column are shown in bold and underlined, respectively. The random-baseline AUPRC equals endpoint activity prevalence in the molecular evaluation set. Molecular predictors are frozen; their performance is unchanged by the revised transcriptomic fitting procedure.

Model	NR-AR	NR-AR-LBD	NR-AhR	NR-Aromat	NR-ER	NR-ER-LBD	NR-PPAR-g	NR macro	SR-ARE	SR-ATAD5	SR-HSE	SR-MMP	SR-p53	SR macro
GAT	0.138	0.087	0.365	0.165	0.263	0.049	0.073	0.163	0.371	0.140	0.129	0.351	0.211	0.240
MACCS + logistic regression	0.696	0.273	0.319	0.206	0.370	0.130	0.057	0.293	0.377	0.086	0.154	0.473	0.292	0.276
RDKit descriptors + XGBoost	0.616	0.341	0.472	0.221	0.307	0.195	0.150	0.329	0.480	0.178	0.154	0.475	0.292	0.316
RDKit descriptors + logistic regression	0.706	0.339	0.445	0.251	0.238	0.194	0.193	0.338	0.470	0.177	0.144	0.552	0.286	0.326
MACCS + XGBoost	0.616	0.374	0.375	0.360	0.384	0.264	0.101	0.353	0.539	0.110	0.111	0.513	0.339	0.323
Morgan + XGBoost	0.764	0.381	0.424	0.370	0.331	0.153	0.246	0.381	0.428	0.159	0.157	0.474	0.247	0.293
Morgan + logistic	0.775	0.231	0.491	0.252	0.309	0.150	0.234	0.349	0.450	0.205	0.226	0.537	0.327	0.349
ChemBERTa	0.744	0.367	0.450	0.283	0.415	0.161	0.044	0.352	0.504	0.180	0.203	0.562	0.301	0.350
Chemprop	0.734	0.232	0.373	0.251	0.335	0.121	0.308	0.336	0.432	0.188	0.313	0.533	0.397	0.373
GCN	0.736	0.306	0.432	0.237	0.388	0.167	0.103	0.339	0.500	0.116	0.383	0.572	0.390	0.392
GraphSAGE	0.802	0.342	0.448	0.350	0.376	0.183	0.074	0.368	0.508	0.131	0.321	0.544	0.431	0.387
GIN	0.818	0.370	0.512	0.309	0.386	0.166	0.072	0.376	0.569	0.212	0.456	0.525	0.436	0.440
Task-specific K=4 arithmetic mean	0.821	0.415	0.574	0.444	0.460	0.219	0.322	0.465	0.618	0.341	0.432	0.622	0.527	0.508
Random baseline	0.038	0.040	0.159	0.074	0.140	0.046	0.045	0.077	0.279	0.058	0.089	0.200	0.136	0.152